

Sources of Long-Term Predictability

Atmosphere

- Madden-Julian Oscillation
- Stratospheric conditions
- Deviations in angular momentum

Land

- Soil moisture
- Snow cover
- Vegetation
- Groundwater

Ocean

- Modes of variability (ENSO, PDO, AMV, etc.)
- Pattern of ocean heat content (surface + depth)
- Sea ice

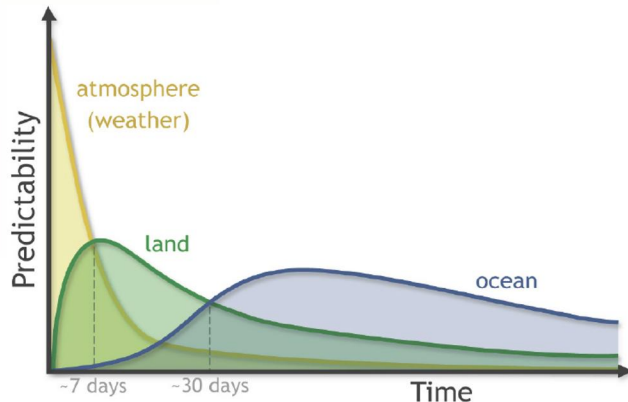


FIGURE: Relative contributions to Earth System predictability from the atmosphere, land, and ocean at different timescales.

Workshop speaker Sarah Gille argued that the ocean curve should be of comparable amplitude to the land curve at timescale for 1-2 weeks. SOURCE: P. Dirmeyer, GMU.